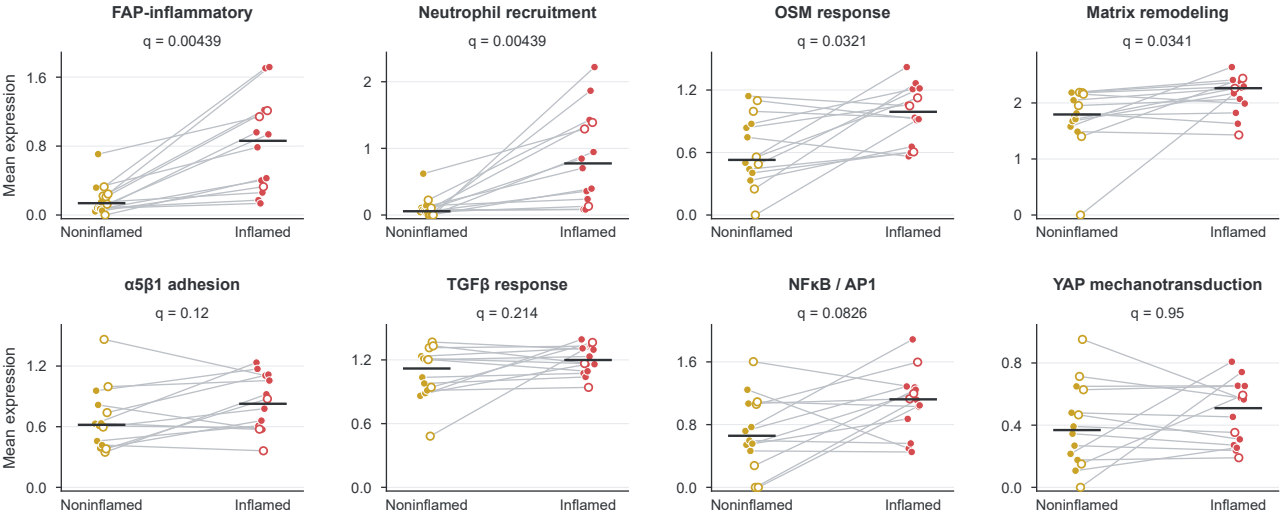


a Paired stromal expression of fibroblast-associated programs

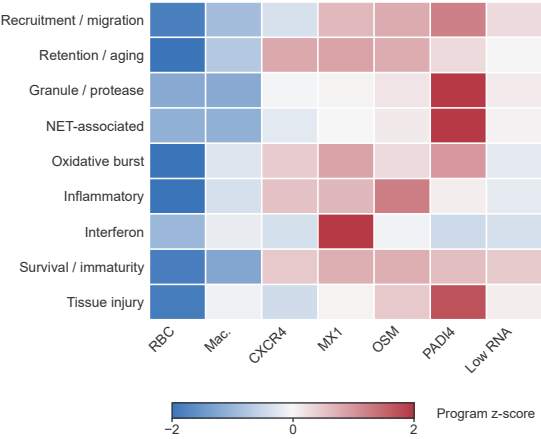
Matched noninflamed and inflamed UC tissue; 14 patients



Paired Wilcoxon q (BH, 8 programs). Lines: patients; bars: medians; open points: <10 recovered cells.

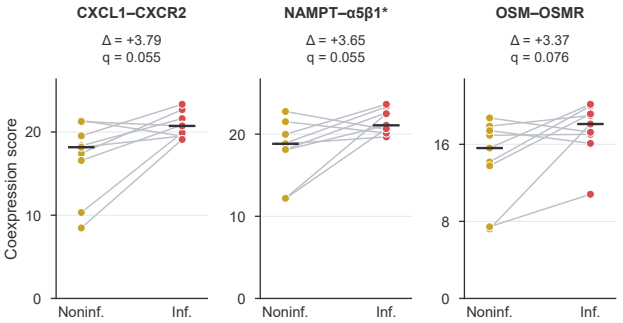
b Pooled RNA programs and annotation controls

All conditions; original gene modules



c Paired ligand-receptor coexpression

9 matched UC patient pairs



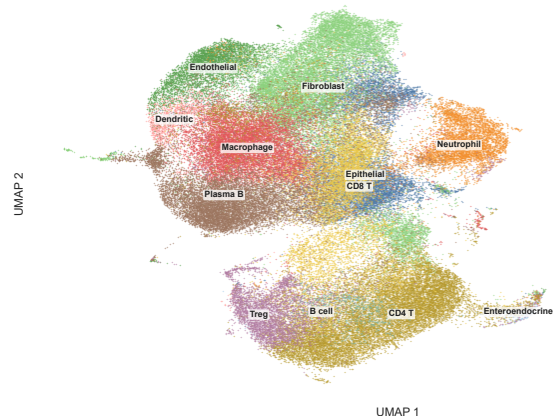
Paired Wilcoxon q (BH, 5 original routes).

*Receptor summary: min(ITGA5, ITGB1).

Δ : median paired change; lines: patients.

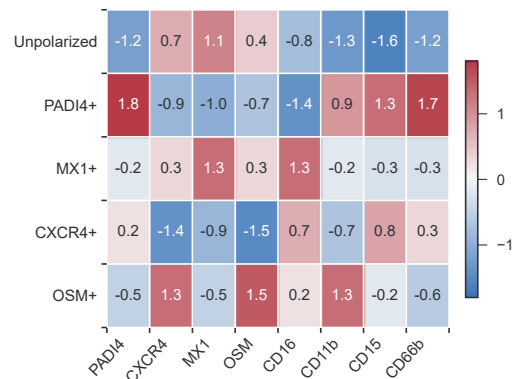
a Broad CODEX cell atlas

245,020 cells; 12 broad annotations



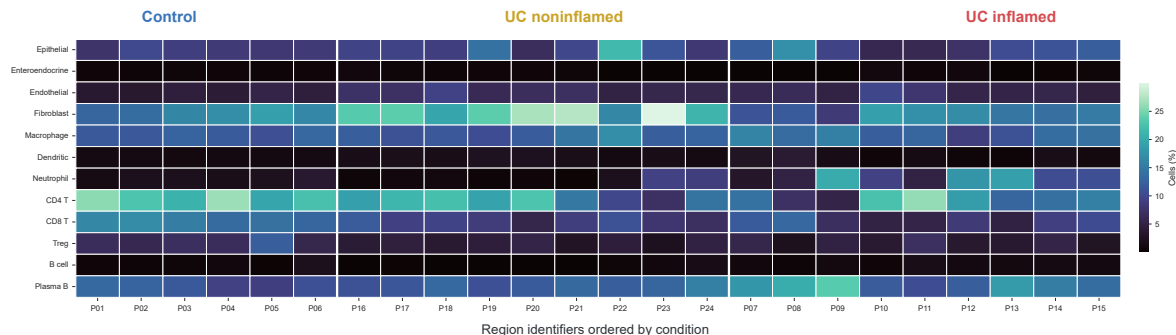
b Neutrophil phenotype validation

Median marker expression, z-scored across states



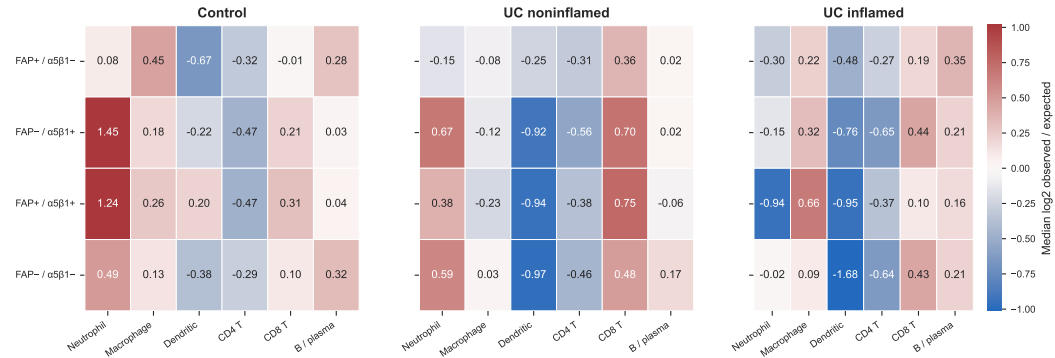
c Region-resolved cell composition

Six control, nine noninflamed UC and nine inflamed UC region identifiers



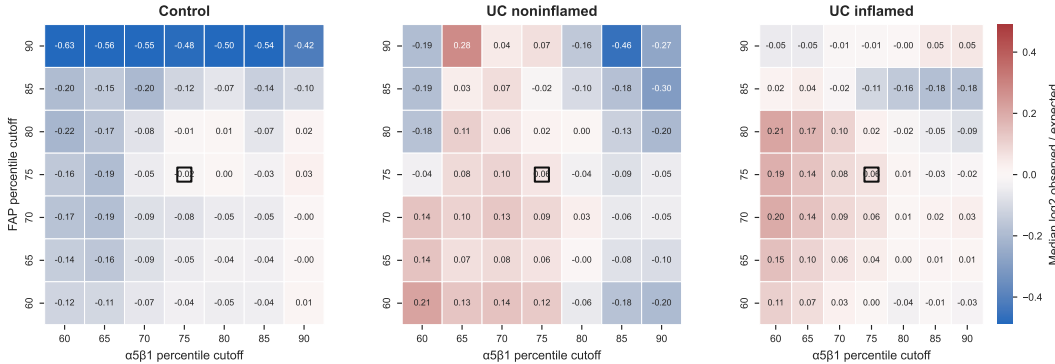
a Immune-lineage spatial specificity

Median log2 observed/expected proximity across FAP/ $\alpha 5\beta 1$ strata



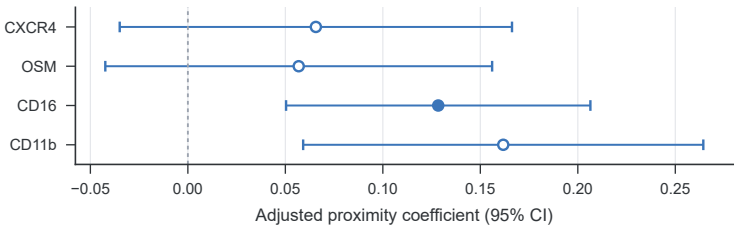
b Threshold sensitivity

FAP and $\alpha 5\beta 1$ percentile cutoffs across tissue conditions



c Adjusted neutrophil proximity effects

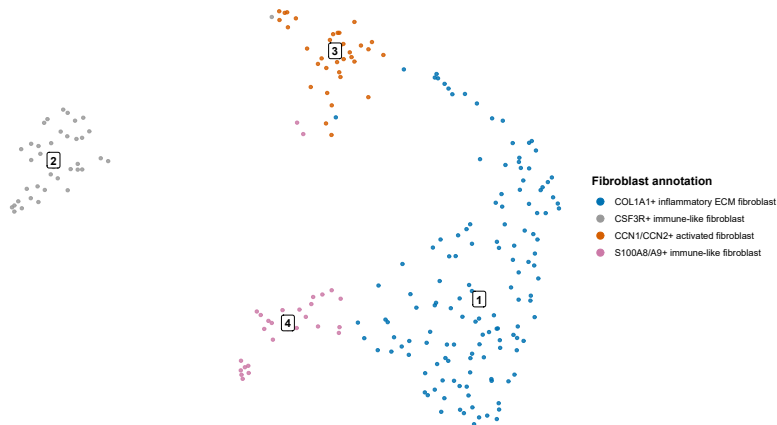
Filled point: restricted spatial-null $q < 0.05$; bars show 95% CI



Model q	Spatial-null q
0.242	0.844
0.242	0.843
0.008	0.032
0.008	0.144

a Fibroblast subclusters and annotations

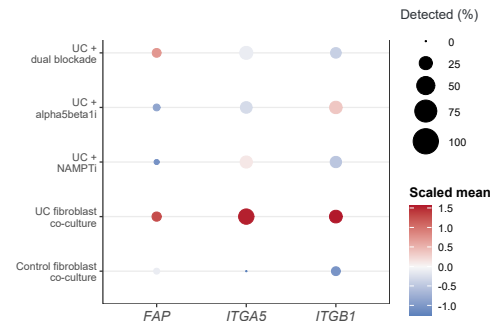
Donor-harmonized fibroblast embedding; annotation resolution 0.4



Immune-like annotations are retained with low confidence because their stromal programs are weak.

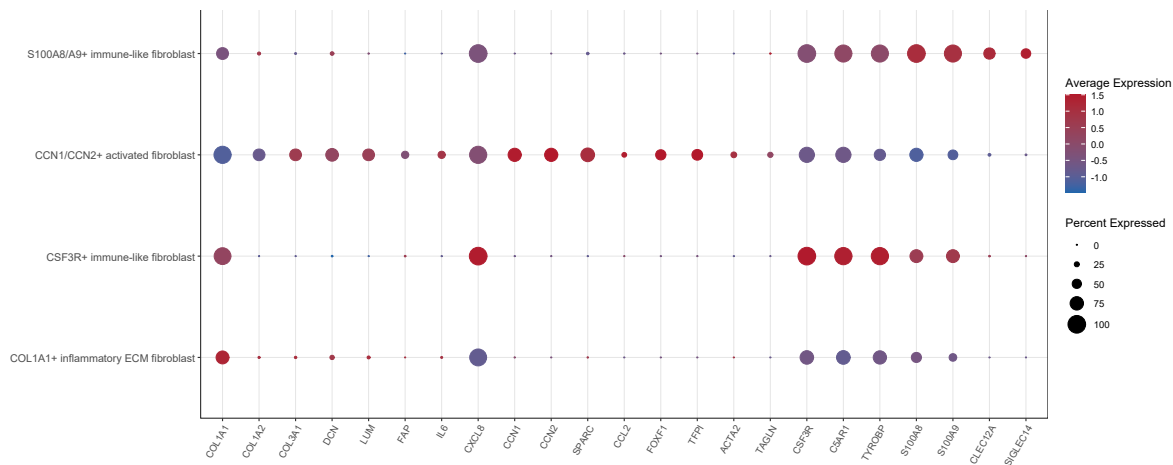
b Target transcripts by condition

FAP, ITGA5 and ITGB1 expression



c Marker programs supporting annotations

Dot size: percent expressed; color: scaled mean expression



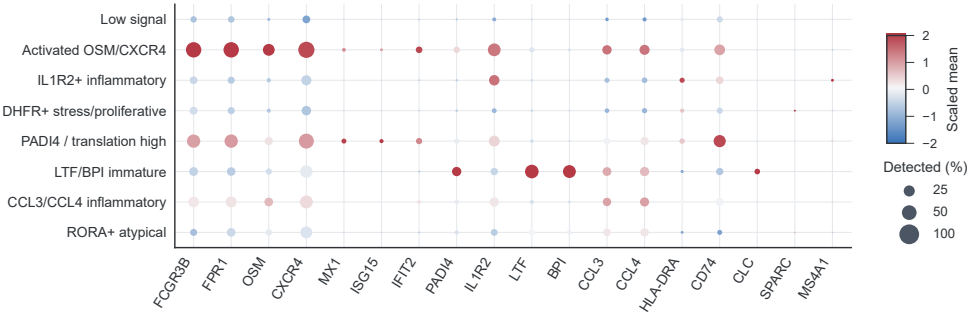
a Neutrophil subcluster atlas

Retained cells colored by transcriptomic subcluster annotation



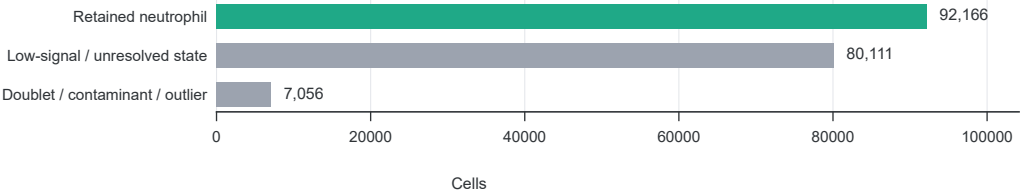
b Marker support for subcluster annotations

Dot size: detected cells; color: scaled mean expression



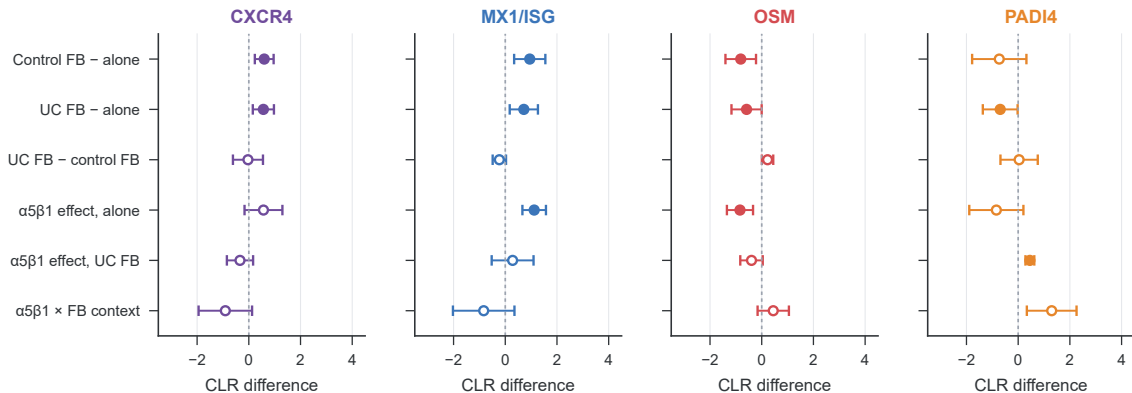
c Cell-retention and cleanup inventory

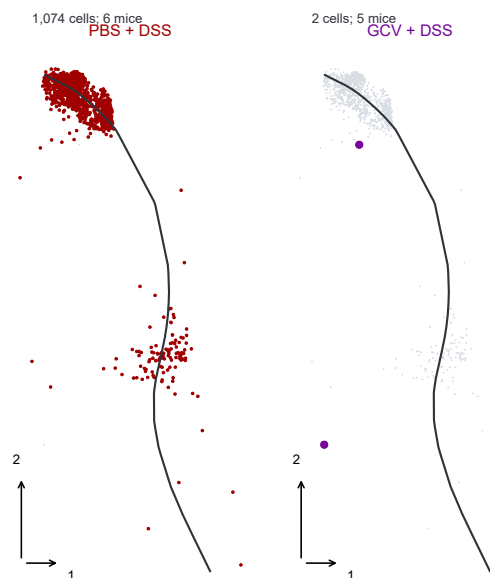
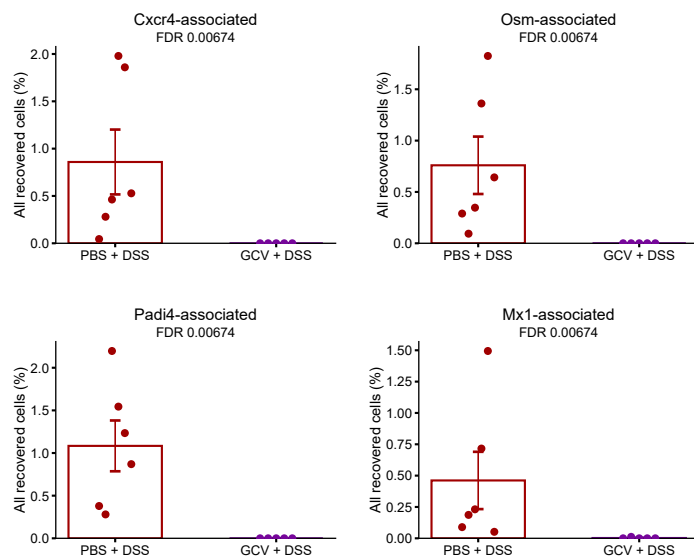
Original counts summarized by disposition



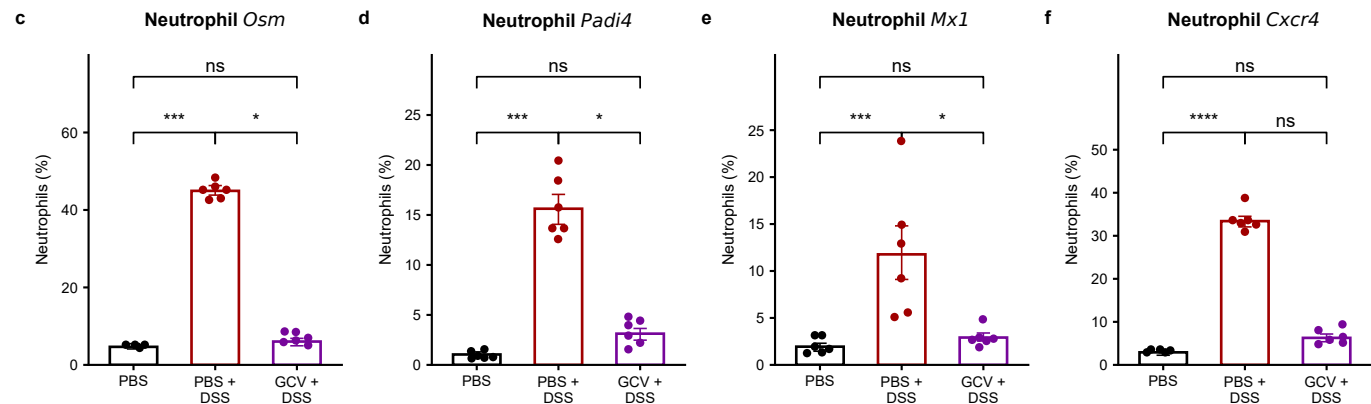
a Donor-aware neutrophil-state contrasts

Six recorded donors; filled points denote BH $q < 0.05$; bars show 95% CI



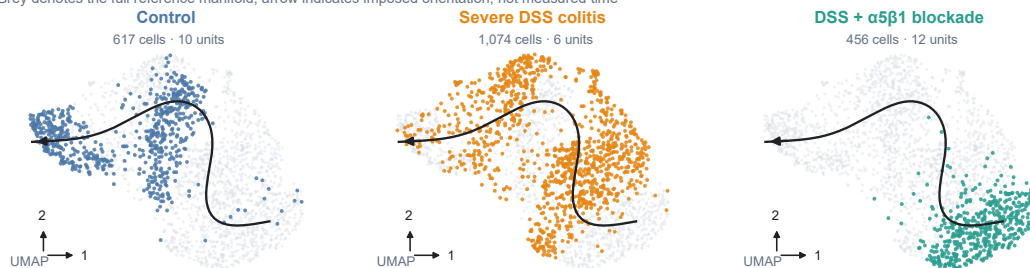
a Neutrophil occupancy after FAP ablation**b Recovered neutrophil-state fractions**

Only two postablation neutrophils: occupancy loss, not a postablation trajectory or within-state estimate.

c-f Neutrophil marker readouts supporting Figure 4

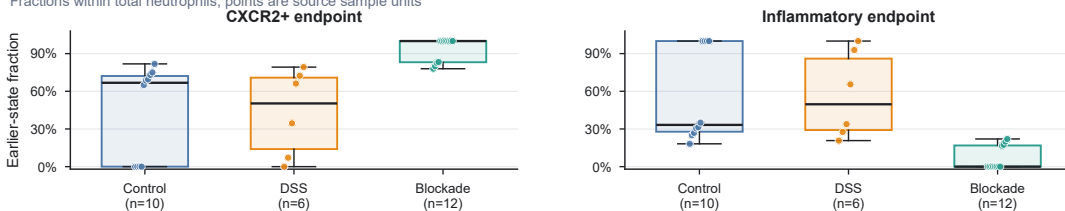
a α5β1 blockade redistributes occupancy along a shared neutrophil continuum

Grey denotes the full reference manifold; arrow indicates imposed orientation, not measured time



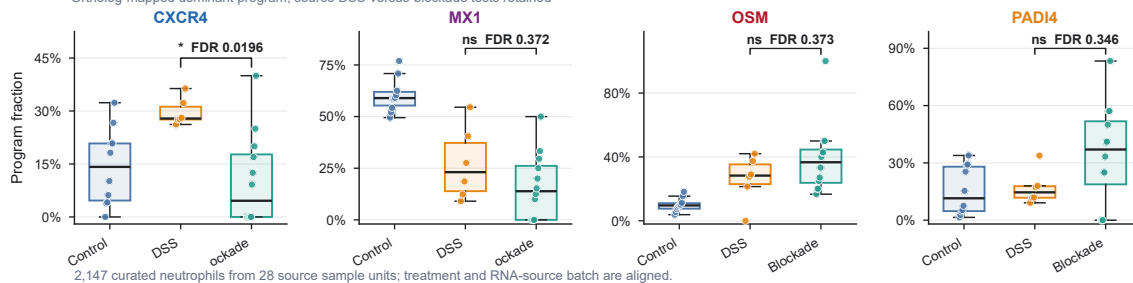
b Endpoint-state redistribution

Fractions within total neutrophils; points are source sample units



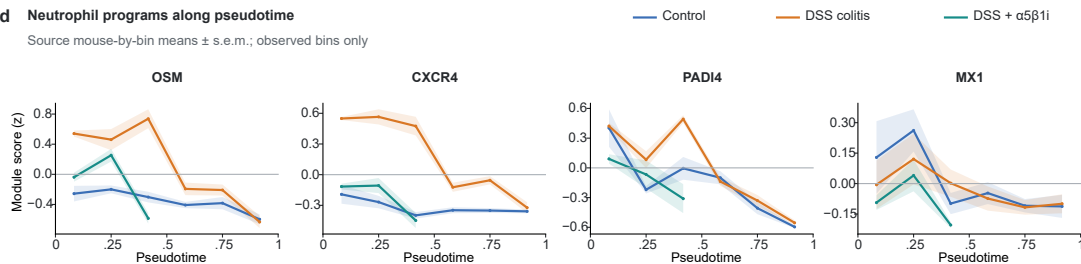
c Transferred human-associated neutrophil programs

Ortholog-mapped dominant program; source DSS-versus-blockade tests retained



d Neutrophil programs along pseudotime

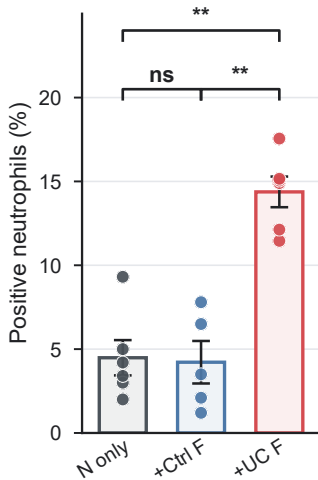
Source mouse-by-bin means ± s.e.m.; observed bins only



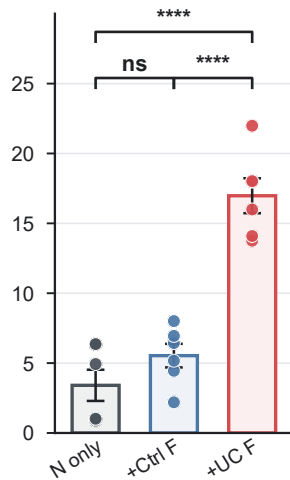
Human fibroblast coculture: neutrophil protein readouts

Neutrophils only + control fibroblasts + UC fibroblasts

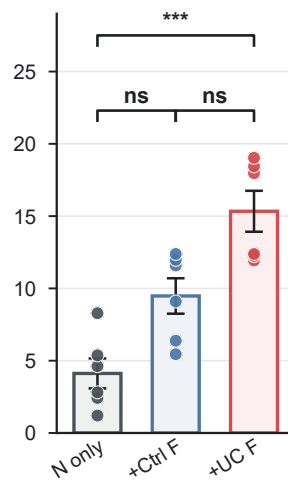
a PADI4+ neutrophils



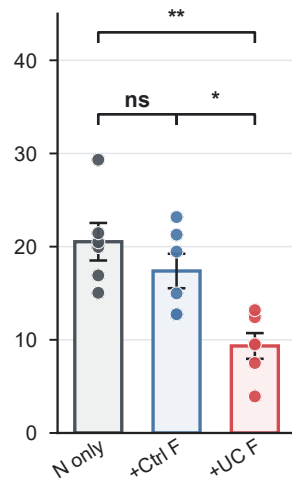
b OSM+ neutrophils



c MX1+ neutrophils

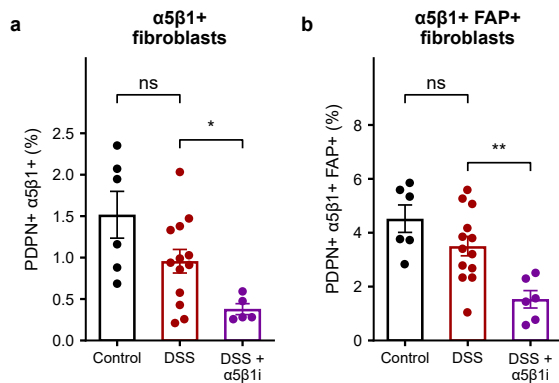


d CXCR4+ neutrophils

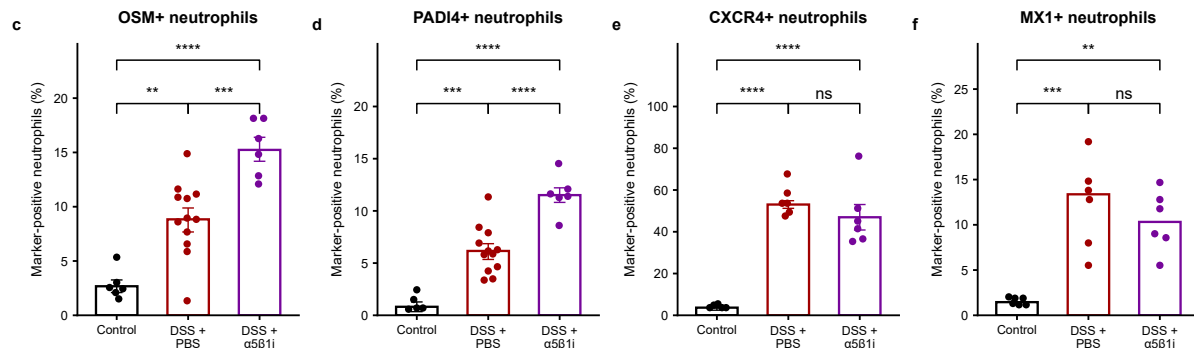


DSS colitis: stromal and neutrophil protein readouts after $\alpha 5\beta 1$ blockade

a,b Stromal $\alpha 5\beta 1$ detection



c-f Marker-positive neutrophils



g-i MPO within marker-positive neutrophils

